HEQIAO RUAN

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EDUCATION

University of Pittsburgh *Ph.D.* *Aug 2022 - Sep 2026*
Thesis:"Understanding Out-of-Distribution: Hypothesis Testing and Causal Inference"
University of California, Davis *M.S.* *Jun 2019*
University of Science and Technology of China *B.S.* *Jul 2017*

SUMMARY

Data Scientist & **Applied Scientist** with a Ph.D. in Statistics and 2.5 years of full-time industry experience prior to doctoral studies, complemented by four years of applied research in statistical ML and causal inference during Ph.D. Specialize in ML modeling, causal reasoning, experimentation, model monitoring, LLM workflows and metrics & feature attributions. I have a proven track record of translating rigorous methods into production-scale product decisions—spanning pricing, incentives, risk management, and trust & safety—and collaborating effectively with product, engineering, and policy teams to deliver measurable business outcomes.
Research Focus: *Statistical Learning, Out-of-Distribution, Streaming Data Analysis, Causal Inference, Large Language Model(LLM) Evaluation & Monitoring & Inference*

SKILLS

Programming: Python(NumPy, Pandas, Scikit-learn,PyTorch),SQL,R,MATLAB, Tableau, PowerBI
LLM: Prompt Engineering, Context Engineering, GraphRAG, Fine-Tuning, Agent Reasoning, LLM Inference, LLM Fine-Tuning, Multi-modal Learning, Acceleration(vLLM, TensorRT-LLM), LLM Evaluation & Monitoring
Statistics & Machine Learning: Hypothesis Testing, Online Learning, Deep Neural Networks, Anomaly Detection, Streaming Data, Causal Meta-Learners, Sequential Testing, Missing Data, Uplift Modeling, A/B Testing, Bayesian Methods, Time-Series Forecasting
AI Tools & Big Data Frameworks: LangChain, LangGraph, PySpark, Hadoop, Docker, Git, AWS

RESEARCH EXPERIENCE

University of Pittsburgh Applied Researcher, Machine Learning *Jun 2023 - Oct 2026*
Developed efficient, scalable, theory-grounded methods for closed-loop distribution-shift detection, attribution, and adaptation in static, streaming, and mini-batch regimes—powering real-time monitoring, proactive alerting, periodic updates, and evaluation-credibility auditing for deployed ML/LLM systems.

Proposed the OOD uncertainty-quantification pipeline and pioneered meta-learner-based concept-drift quantification; validated on LLM monitoring, multimodal attribution, and production ML, with direct transfer to LLM evaluation via attribution of benchmark-score fluctuations and degradation-induced task failures.

Random Forest Based Statistical Drift Detection-Model-Agnostic, Multi-Modal

- Developed a random-forest-based permutation test for detecting distribution shift in LLM inference pipelines; effectively distinguishes true semantic and prompt distribution changes from stochastic noise, and reduces computational latency via parallelization, enabling real-time alerts for prompt drift and model degradation.
- Designed and implemented RFPerm-as-Judge, a real-time streaming distributional alignment framework that evaluates **LLM-synthetic data** against observational benchmarks, enabling quality control in A/B test cold starts and data augmentation while reducing distributional bias risk from synthetic data.
- Built an online distribution shift adaptation pipeline that incorporates online anomaly detection scores, pseudo-outcome risk, and rolling statistics as **Clever Covariates** into an optional correction mechanism for LLM inference. Validated on **LLM-generated text** and multi-modal outputs for mitigating reasoning-path and generation-quality decay, with performance decay attributed to distributional change dimensions.

Real-Time Feature Selection for Distribution Shift via Meta-Learners

- Proposed **out-of-distribution (OOD) variable importance** via a pseudo-outcome learner with permutation variable importance, enabled interpretable attribution and quantification of distribution shift drivers.
- Extended the variable-importance framework feature selection regime, proposed **feature selection for distribution shift (FSDS)**, leveraged meta-learners for concept drift and random forest domain classifiers with MMD for covariate shift, achieving state-of-the-art attribution and real-time feature-level interpretability. Further extended the attribution tool to sentence- and image-region-level attribution via post-hoc subset localization, enabling **multimodal attribution** at text token, image patch, and video frame levels.
- Extended FSDS to a horizontal federated aggregation framework with inverse-variance weighting that aligns block-level out-of-distribution(OOD) variable-importance across sites with 0.9 consistency, keeps local updates on-device, and reduces communication cost by 90%.
- Built a **continuous feature-relevance monitoring system** based on permutation importance, tracking variable-importance evolution via rolling windows to compress and pinpoint removal-decision diagnostic time and enable timely retirement of stale features via feature-quality decay monitoring in **ML pipelines**.
- Developed **post-hoc subset localization**, a calibrated layer separating genuine biomarker signal from shifts inherited through feature dependence, integrated into rolling-window monitoring to flag stale or inherited regions; demonstrated on RNA-seq, recovering expected gating markers and cell-type signatures.

Online Distribution Shift Adaptation System — Meta-Learners, Multi-Modal

- Extended RFPerm to streaming **LLM inference** with p-value-based predictive profiling for real-time performance decay localization and data-driven update timing; further extended the **meta-learner-as-distance-estimator** pipeline to streaming batches, combining batch-level pseudo-outcome risk with online anomaly detection to pinpoint concept drift onset and attribute performance decay to distributional shift.
- Implemented **OOD uncertainty quantification** on pseudo-outcome risk via **online bootstrap** for **LLM inference**, guided layer-wise training/freezing strategies in the deployed neural networks, measured distributional misalignment risk of generated content and reasoning paths; combined with online PFI monitoring and drift detection to form a closed loop for monitoring and updates, supporting prompt/embedding attribution, multi-level metric characterization, and continuous subgroup localization.
- Designed and implemented a multi-armed bandit algorithm with adaptive exploration for online model fusion, adopting the feature selection(**FSDS**) results as weights into the next model iteration; while improving allocation efficiency, it enhances the stability of **LLM inference** under distribution shift.

WORK EXPERIENCE

Suning R&D Center USA(California) Data Scientist *Sep 2019 - Dec 2021*

Built data-driven solutions to boost product growth and control financial risk in Suning E-commerce Platform.

End-to-End ML Model: Risk Modeling & Product Growth

- Designed risk-aware modeling strategies integrating fraud prevention and customer growth for large-scale e-commerce financial products. Integrated merchant transaction fraud signals into credit approval workflows, expanding eligible customer coverage for consumer loan by 450% while maintaining portfolio risk constraints.
- Engineered 300+ behavioral and transactional features(browsing trajectories, IP profiling, purchasing history), improved AUC by 0.04 through **XGBoost** and generating approximately $4M in incremental profit.

Model Monitoring, Interpretability & Governance

- Built a streamlined model monitoring pipeline covering feature quality checks, drift detection, performance degradation interpretability, and retraining signals across production models. Implemented **Shapley-value-based explanations** to support compliance review, operational debugging, and stakeholder trust.
- Designed streamlined, automated reason-code generation for case investigations, reduced manual review effort by $\sim$ 1500 person-days. Submitted one patent application.

Anti-Money Laundering Project - From Zero to One

- Served as a core contributor to Suning's first **AML** detection system from scratch, co-designed anomaly-detection models via **Isolation Forest** to identify suspicious accounts and transactions.

- Improved detection rates by 300%, prevented $2M potential losses, significantly reduced regulatory risk.

TikTok.Inc (CA) Data Scientist Intern Platform Generating Content Team *Jun 2025 - Sep 2025*

Creator Reward Program - Uplift and Elasticity Modeling

- Led causal analysis to quantify creator reward sensitivity and downstream impact on content quality and user engagement. Build uplift models using **DoubleML** to estimate individual treatment effects and identify **three key business factors** adopted into the reward pricing formula, guided incentive pricing strategies.
- Designed and implemented causal mediation frameworks (**BART, DeepMed**) to disentangle direct and indirect effects of rewards on creator–consumer interactions, translating results into deployable decision rules. Validated the causal findings through online experiments and integrate insights into live systems.

Metrics Attribution Analysis in Content Ecosystem

- Developed meta-learner-based testing and feature selection methods for feature-store change attribution and metric root-cause analysis, improving interpretability; presented at a company-wise symposium.

PROJECTS

Early-Warning Signal Detection for Battery Thermal-Runaway *Jun 2024 - July 2026*

- Developed a slope-based online abrupt change detection framework that monitors battery time series in real-time, enabled early warning of thermal runaway precursors with 20-second lead time(Precision: 85%).
- Designed a neural network integrated Transformer and iTransformer with a Mixture-of-Experts(MoE) mechanism for online thermal anomaly detection in battery systems, reducing the missed-detection rate to 1.3%.

Single-cell Trajectory Inference, implementation and modification *Jan 2018 - Jun 2018*

- Implemented multiple trajectory inference methods(TSCAN, Mpath) in **R** to infer the order of cells along developmental trajectories. Modified existing methodology by adding **TSNE, ICA, GAT(Graph Attention Network)** for dimensionality reduction and **K-means, DBSCAN** for clustering.
- Achieved improved accuracy in modified **TSCAN** by **20%** and improved robustness under perturbations by **10%**, identified genes related to cell differentiation confirmed by previous publications.

SELECTED PUBLICATIONS & PRESENTATIONS

High Prevalence of Diabetic Retinopathy in an Outpatient Podiatry Clinic and Associated Barriers to Ophthalmic Care(2025), Larimer-Picciani, R.Brown, Heqiao Ruan et.al. Clinical Ophthalmology

Online distribution shift detection in ever-evolving data streams via Random Forest and Pseudo-Outcome Risk (2026+), Heqiao Ruan & Lucas Mentch

Leveraging Causal Infernce for Detecting Distribution Shift Robust & Efficient Tests for Distribution Shift via Random Forests(2026+), Ruan & Mentch

Leveraging Meta-Learner for Detecting Distribution Shift(2026+), Ruan & Mentch

Disentangling Concept Drift and Covariate Shift via R-learner(2026+), Ruan & Mentch

Hypothesis Test for Distribution Shift via Random Forest, ASA Banquet, Apr 2025, Pittsburgh, PA

Leveraging Causal Inference for Detecting Distribution Shift with Application to Metrics Attribution Analysis, Heqiao Ruan & Jian Tian & Lucas Mentch, **Tiktok Ph.D. Intern Expo,**, San Jose, CA. Jul 2025

Data-Driven Detection for Thermal Runaway, **MDS-Rely** Spring Meeting, Apr 2026

Review: NIPS workshop 2026

REFERENCES

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